

## CAPE OTWAY LH, Australia

WMO: 948420

Lat: 38.8551S

Lon: 143.5140E

Elev: 272

StdP: 14.55

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 43.1       | 44.5 | 31.2                          | 25.7 | 57.2 | 33.6 | 28.5 | 54.1 | 37.0                  | 49.9 | 33.4 | 50.5 | 11.1                  | 0    | 0.673 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 10.0                   | 86.4            | 63.5 | 81.3 | 62.1 | 76.4 | 61.2 | 67.2                | 76.9 | 65.6 | 74.1 | 64.3 | 71.2 | 15.5                 | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.6                            | 92.2 | 68.9 | 63.2 | 87.5 | 67.8 | 61.8 | 83.4 | 66.6 | 31.8          | 76.8 | 30.6 | 74.5 | 29.6 | 71.5 | 85.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 31.7          | 28.7 | 26.0 | DB                         | 40.0 | 98.9               | 1.2 | 4.9                                                | 39.1 | 102.5      | 38.3 | 105.3      | 37.6 | 108.1      | 36.7 | 111.7 | (4) |
| (5)               |      |      | WB                         | 37.7 | 73.1               | 1.3 | 4.2                                                | 36.8 | 76.2       | 36.0 | 78.7       | 35.3 | 81.1       | 34.4 | 84.1  | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 57.1   | 63.3  | 64.3  | 62.0  | 58.6  | 55.4  | 52.3  | 51.1  | 51.6  | 53.4  | 55.6  | 57.8  | 60.5  |  |  |
|                                                                        | DBStd       | 6.46   | 5.81  | 5.74  | 5.47  | 4.67  | 3.75  | 2.87  | 2.75  | 3.32  | 4.30  | 5.39  | 5.36  | 5.62  |  |  |
|                                                                        | HDD50       | 55     | 0     | 0     | 0     | 0     | 1     | 9     | 19    | 17    | 7     | 2     | 0     | 0     |  |  |
|                                                                        | HDD65       | 3087   | 103   | 71    | 127   | 204   | 299   | 382   | 432   | 415   | 351   | 302   | 231   | 170   |  |  |
|                                                                        | CDD50       | 2652   | 412   | 400   | 373   | 259   | 169   | 77    | 52    | 67    | 109   | 175   | 235   | 324   |  |  |
|                                                                        | CDD65       | 207    | 50    | 51    | 35    | 12    | 2     | 0     | 0     | 0     | 2     | 10    | 16    | 29    |  |  |
|                                                                        | CDH74       | 1569   | 398   | 350   | 249   | 60    | 1     | 0     | 0     | 0     | 8     | 98    | 138   | 267   |  |  |
|                                                                        | CDH80       | 583    | 169   | 144   | 92    | 12    | 0     | 0     | 0     | 0     | 0     | 23    | 34    | 109   |  |  |
| (14) Wind                                                              | WSAvg       | 14.0   | 13.0  | 12.9  | 13.0  | 12.8  | 13.8  | 13.8  | 15.2  | 15.5  | 15.5  | 14.6  | 13.9  | 14.0  |  |  |
| (15) Precipitation                                                     | PrecAvg     | 37.6   | 1.8   | 1.6   | 2.3   | 2.9   | 3.6   | 3.9   | 4.6   | 4.6   | 3.7   | 3.3   | 2.8   | 2.2   |  |  |
|                                                                        | PrecMax     | 50.9   | 4.8   | 6.1   | 5.5   | 7.5   | 8.8   | 7.7   | 8.1   | 8.1   | 6.9   | 6.7   | 5.5   | 4.8   |  |  |
|                                                                        | PrecMin     | 25.0   | 0.3   | 0.1   | 0.5   | 0.6   | 0.9   | 1.3   | 1.8   | 1.5   | 1.2   | 0.3   | 0.6   | 0.3   |  |  |
|                                                                        | PrecStd     | 5.6    | 0.9   | 1.3   | 1.1   | 1.6   | 1.7   | 1.4   | 1.7   | 1.5   | 1.4   | 1.5   | 1.2   | 1.1   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 93.4  | 91.7  | 89.4  | 80.9  | 70.1  | 63.6  | 61.4  | 67.7  | 74.6  | 83.4  | 85.1  | 91.6  |  |  |
|                                                                        |             | MCWB   | 66.6  | 65.1  | 64.2  | 61.3  | 56.1  | 54.5  | 52.4  | 53.7  | 58.1  | 59.2  | 62.2  | 64.2  |  |  |
|                                                                        | 2%          | DB     | 84.2  | 84.2  | 81.0  | 74.1  | 66.8  | 60.2  | 58.7  | 62.7  | 69.1  | 76.0  | 77.7  | 81.6  |  |  |
|                                                                        |             | MCWB   | 63.8  | 63.1  | 62.9  | 58.6  | 55.1  | 53.1  | 50.8  | 52.0  | 55.1  | 57.7  | 60.4  | 61.3  |  |  |
|                                                                        | 5%          | DB     | 77.1  | 77.0  | 74.5  | 69.1  | 63.8  | 58.1  | 56.9  | 59.4  | 64.3  | 70.1  | 71.7  | 73.8  |  |  |
|                                                                        |             | MCWB   | 64.0  | 63.8  | 61.3  | 57.8  | 55.3  | 52.4  | 50.2  | 50.6  | 53.6  | 55.7  | 59.5  | 60.8  |  |  |
|                                                                        | 10%         | DB     | 70.4  | 71.0  | 68.8  | 65.0  | 61.1  | 56.6  | 55.4  | 56.8  | 60.1  | 63.7  | 65.5  | 67.4  |  |  |
|                                                                        |             | MCWB   | 63.0  | 63.3  | 61.8  | 57.8  | 54.8  | 51.7  | 49.9  | 50.4  | 51.9  | 54.8  | 59.2  | 60.6  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 69.8  | 69.0  | 67.3  | 64.5  | 59.9  | 56.8  | 55.0  | 56.8  | 59.7  | 62.7  | 66.0  | 67.1  |  |  |
|                                                                        |             | MCDB   | 80.9  | 78.9  | 77.0  | 76.6  | 64.3  | 59.3  | 57.9  | 64.4  | 68.7  | 74.3  | 76.5  | 79.8  |  |  |
|                                                                        | 2%          | WB     | 67.1  | 66.8  | 65.4  | 61.7  | 58.0  | 54.9  | 53.1  | 54.0  | 56.6  | 59.4  | 63.0  | 64.6  |  |  |
|                                                                        |             | MCDB   | 77.7  | 74.8  | 73.7  | 67.3  | 62.5  | 58.0  | 56.2  | 58.8  | 65.4  | 70.0  | 71.5  | 75.1  |  |  |
|                                                                        | 5%          | WB     | 65.1  | 65.5  | 63.8  | 60.2  | 56.8  | 53.7  | 51.9  | 52.6  | 54.8  | 57.5  | 61.1  | 62.5  |  |  |
|                                                                        |             | MCDB   | 73.9  | 71.9  | 70.2  | 65.2  | 61.3  | 56.9  | 55.0  | 56.9  | 60.9  | 66.1  | 68.4  | 70.1  |  |  |
|                                                                        | 10%         | WB     | 63.3  | 64.2  | 62.4  | 58.9  | 55.7  | 52.6  | 51.0  | 51.3  | 53.3  | 55.7  | 59.2  | 60.8  |  |  |
|                                                                        |             | MCDB   | 69.8  | 69.5  | 68.0  | 63.5  | 60.0  | 55.9  | 54.3  | 55.5  | 58.6  | 62.1  | 65.0  | 66.7  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.2  | 10.0  | 9.4   | 8.8   | 7.5   | 6.9   | 7.1   | 8.1   | 9.4   | 10.7  | 10.1  | 10.7  |  |  |
|                                                                        |             | MCDBR  | 24.3  | 21.4  | 20.1  | 16.0  | 11.6  | 8.5   | 8.6   | 12.5  | 16.7  | 22.3  | 22.3  | 23.6  |  |  |
|                                                                        |             | MCWBR  | 9.4   | 8.4   | 8.6   | 8.2   | 6.8   | 5.9   | 5.7   | 6.9   | 8.6   | 10.0  | 10.0  | 10.0  |  |  |
|                                                                        | 5% WB       | MCDBR  | 19.2  | 13.9  | 14.9  | 11.7  | 9.2   | 7.2   | 6.8   | 9.9   | 13.5  | 17.7  | 17.6  | 17.8  |  |  |
|                                                                        |             | MCWBR  | 9.1   | 7.5   | 7.9   | 7.5   | 6.4   | 5.8   | 5.7   | 6.8   | 8.2   | 9.4   | 9.2   | 9.1   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.349 | 0.348 | 0.334 | 0.326 | 0.311 | 0.309 | 0.303 | 0.320 | 0.337 | 0.340 | 0.345 | 0.342 |  |  |
|                                                                        | taud        |        | 2.531 | 2.544 | 2.585 | 2.583 | 2.606 | 2.591 | 2.610 | 2.561 | 2.503 | 2.509 | 2.516 | 2.543 |  |  |
|                                                                        | Ebn at Noon |        | 312   | 304   | 296   | 278   | 263   | 252   | 263   | 276   | 289   | 303   | 311   | 315   |  |  |
|                                                                        | Edh at Noon |        | 35    | 33    | 29    | 25    | 21    | 20    | 21    | 25    | 31    | 33    | 35    | 35    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2200  | 1898  | 1468  | 1019  | 678   | 532   | 587   | 848   | 1226  | 1666  | 1953  | 2200  |  |  |
|                                                                        | RadStd      |        | 142   | 88    | 77    | 58    | 30    | 37    | 37    | 47    | 71    | 118   | 119   | 142   |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.83 | N/A     | N/A    | N/A     | N/A   | +1.12 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (31 neighbors) |  | +0.88 | N/A     | N/A    | +1.58   | +0.70 | N/A   | -20         | -245  | +317  | +86   |

Nomenclature: See separate page